Exercise 1-2

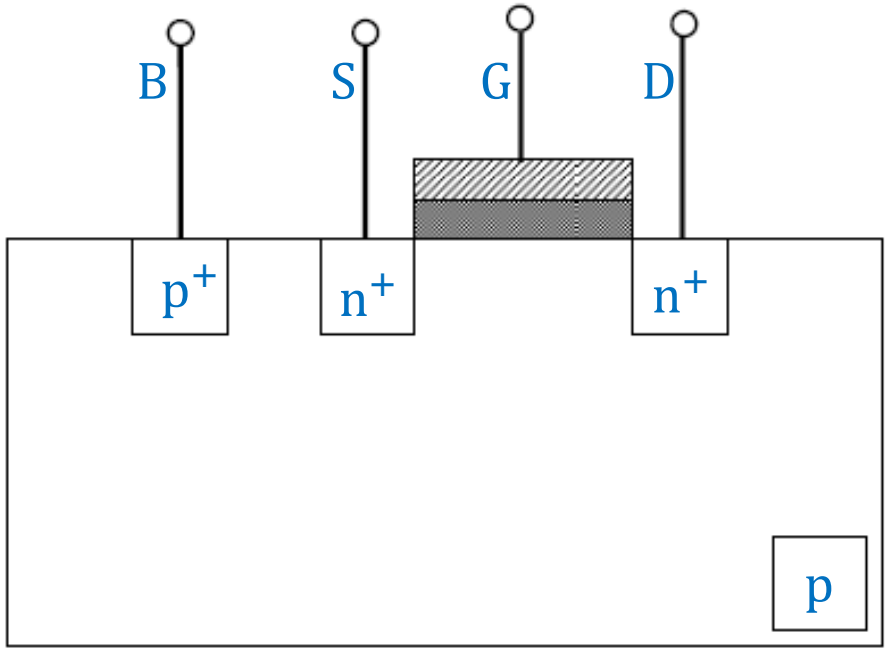
Winter Term 2025/26

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Exercise 1-2 – Task 1.a

1. In **Figure 1**, the cross-section of an **MOS** transistor implemented using a single-well process is shown.

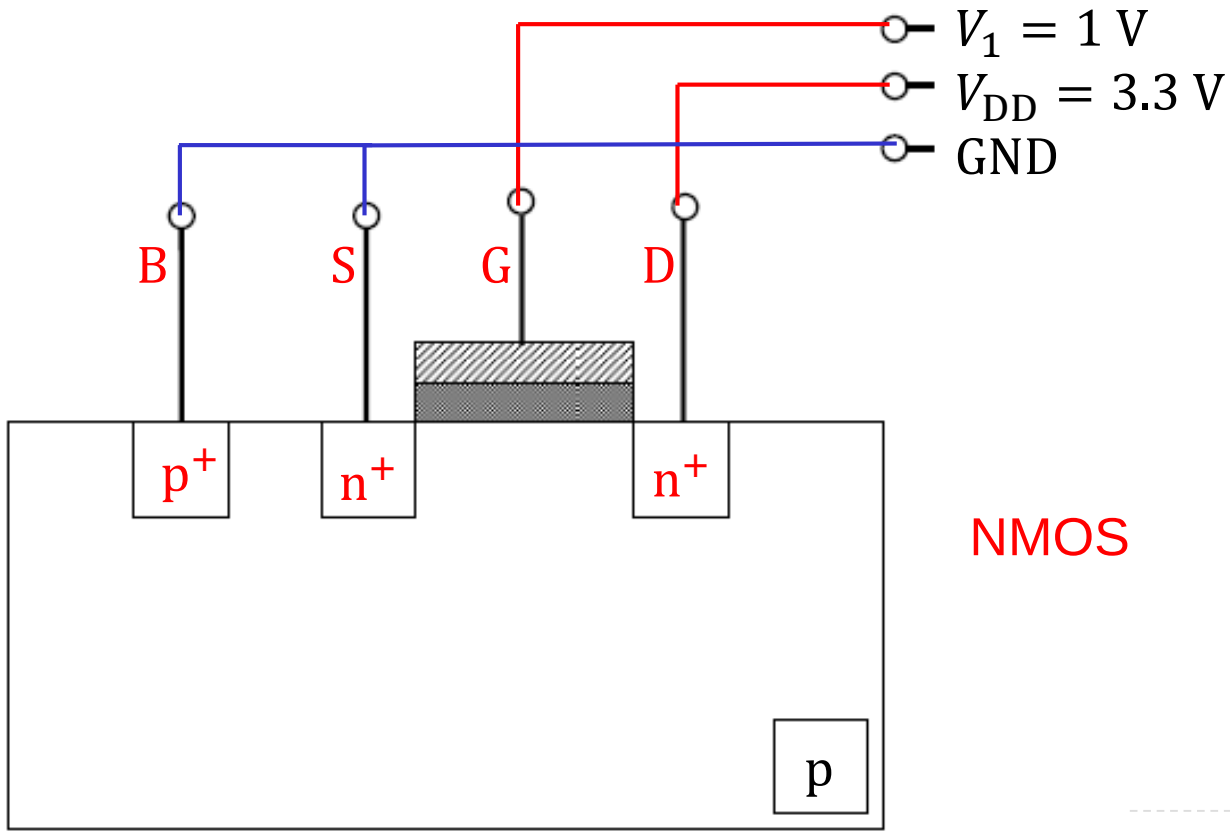
   a. State the type of doping in the respective boxes and label the terminals. Differentiate between weak (n, p) and strong (n^+, p^+) doping concentration. State if it is a PMOS or an NMOS transistor?



NMOS

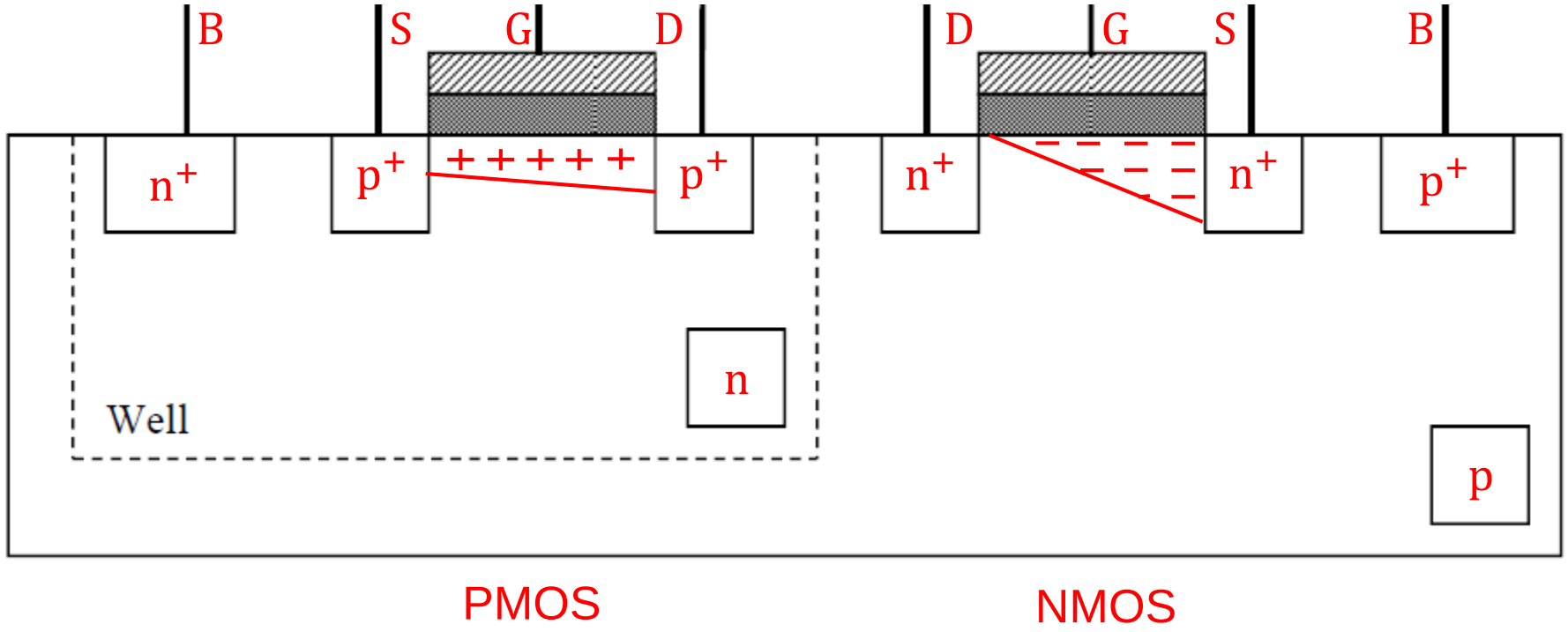
Exercise 1-2 – Task 1.b

- b-Connect each contact of the transistor to one of the voltage sources V_1, V_{DD}, and GND for operating the transistor in the **saturation region**. Note that the threshold voltage V_{th} is $0.5\ \mathrm{V}$.



Exercise 1-2 – Task 1.c

- C-Sketch the cross-section of an NMOS and a PMOS transistor implemented in a double-well process. Sketch the channel shapes assuming the NMOS transistor operates in the saturation and the PMOS transistor in linear region



Exercise 1-2 – Task 2.a

2. Depending on the biasing condition, a MOS transistor operates in a different region.

   a. State the current equation of each region and the corresponding biasing condition!

subthreshold region $v_{GS} < V_{th}$

$i_D \approx 0$

linear region $v_{GS} > V_{th}$ $v_{DS} < v_{GS} - V_{th}$

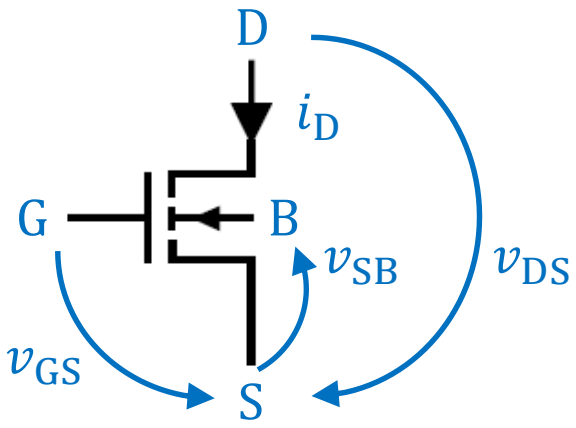
$$i_D = \beta_0 \frac{W}{L}\left(v_{GS} - V_{th} - \frac{v_{DS}}{2}\right) v_{DS}(1 + \lambda v_{DS}) \qquad \lambda \sim \frac{\Delta L}{L}$$

saturation region $v_{GS} > V_{th}$ $v_{DS} > v_{GS} - V_{th}$

$$i_D = \frac{\beta_0}{2}\frac{W}{L}(v_{GS} - V_{th})^2(1 + \lambda v_{DS}) \qquad \text{with} \quad \beta_0 = \mu \frac{\epsilon_{ox}}{t_{ox}}$$

Exercise 1-2 – Task 2.b

- b-State the type of the transistor shown in **Figure 2!** Label the terminals, the currents, and the voltages!



NMOS

Exercise 1-2 – Task 2.c

- C- Calculate the current I_{D} for $\beta_0 = 250\ \mu\mathrm{A/V}^2$, $V_{\mathrm{th}} = 0.2\ \mathrm{V}$, $W = 1\ \mu\mathrm{m}$, $L = 500\ \mathrm{nm}$, $V_{\mathrm{GS}} = 0.5\ \mathrm{V}$, and $V_{\mathrm{DS}} = 0.3\ \mathrm{V}$

$$I_{\mathrm{D,LIN}} = \beta_0 \frac{W}{L}\left(V_{\mathrm{GS}} - V_{\mathrm{th}} - \frac{V_{\mathrm{DS}}}{2}\right)V_{\mathrm{DS}} = 22.5\ \mu\mathrm{A}$$

$$I_{\mathrm{D,SAT}} = \frac{\beta_0}{2}\frac{W}{L}(V_{\mathrm{GS}} - V_{\mathrm{th}})^2 = 22.5\ \mu\mathrm{A}$$

Exercise 1-2 – Task 2.d

d. How much does the drain current i_{D} vary if the gate-source voltage v_{GS} changes by $\pm 100\ \mathrm{mV}$?

Case 1:

$V_{\mathrm{GS}} = 0.5\ \mathrm{V} + 100\ \mathrm{mV} = 0.6\ \mathrm{V}$

$V_{\mathrm{DS}} = 0.3\ \mathrm{V} < V_{\mathrm{GS}} - V_{\mathrm{th}} = 0.4\ \mathrm{V}$ → linear region

$$I_{\mathrm{D,LIN}} = \beta_0 * \frac{W}{L}\left(V_{\mathrm{GS}} - V_{\mathrm{th}} - \frac{V_{\mathrm{DS}}}{2}\right) V_{\mathrm{DS}} = 37.5\ \mu\mathrm{A}$$

$$\Delta i_{\mathrm{D+100\ mV}} = 15\ \mu\mathrm{A}$$

Case 2:

$V_{\mathrm{GS}} = 0.5\ \mathrm{V} - 100\ \mathrm{mV} = 0.6\ \mathrm{V}$

$V_{\mathrm{DS}} = 0.3\ \mathrm{V} > V_{\mathrm{GS}} - V_{\mathrm{th}} = 0.2\ \mathrm{V}$ → saturation region

$$I_{\mathrm{D,SAT}} = \frac{\beta_0}{2}\frac{W}{L}(V_{\mathrm{GS}} - V_{\mathrm{th}})^2 = 10\ \mu\mathrm{A}$$

$$\Delta i_{\mathrm{D-100\ mV}} = -12.5\ \mu\mathrm{A}$$

Exercise 1-2 – Task 3.a+b

3. In **Figure 3**, the current i is monitored while the voltages v is varied.

   a. Sketch the current i versus the voltage v using the parameters given in task 2.c!

   b. Mark the operating region(s) in the sketch!

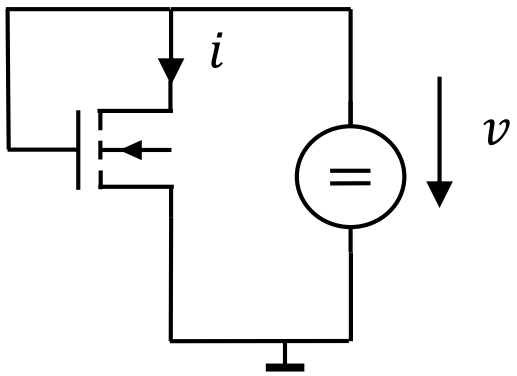
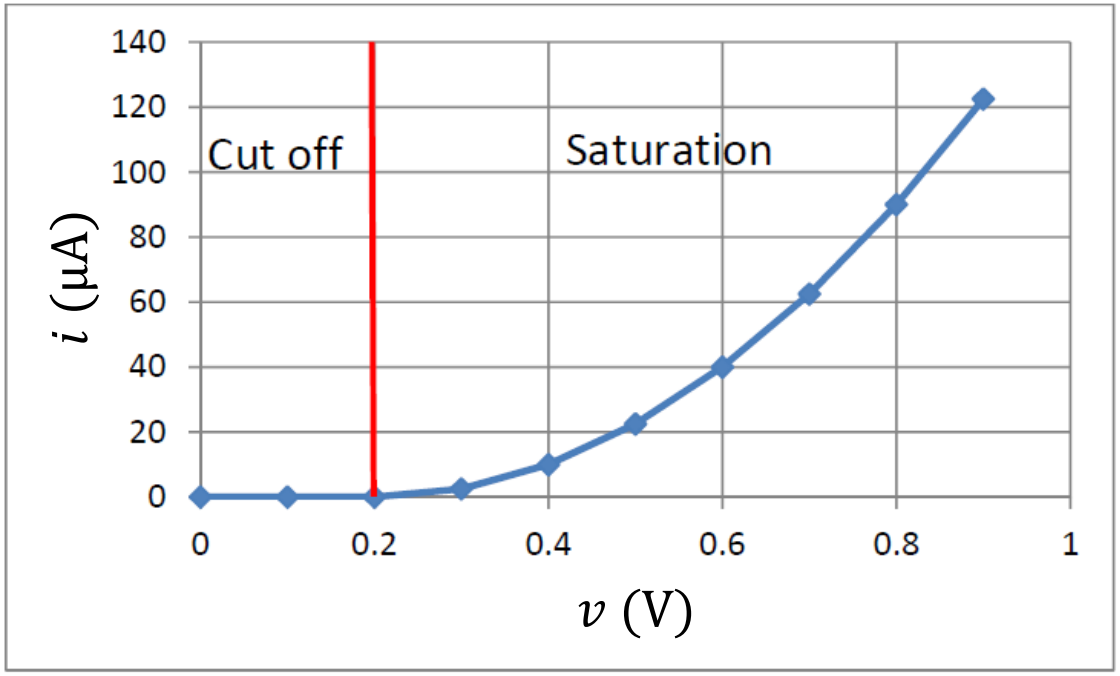


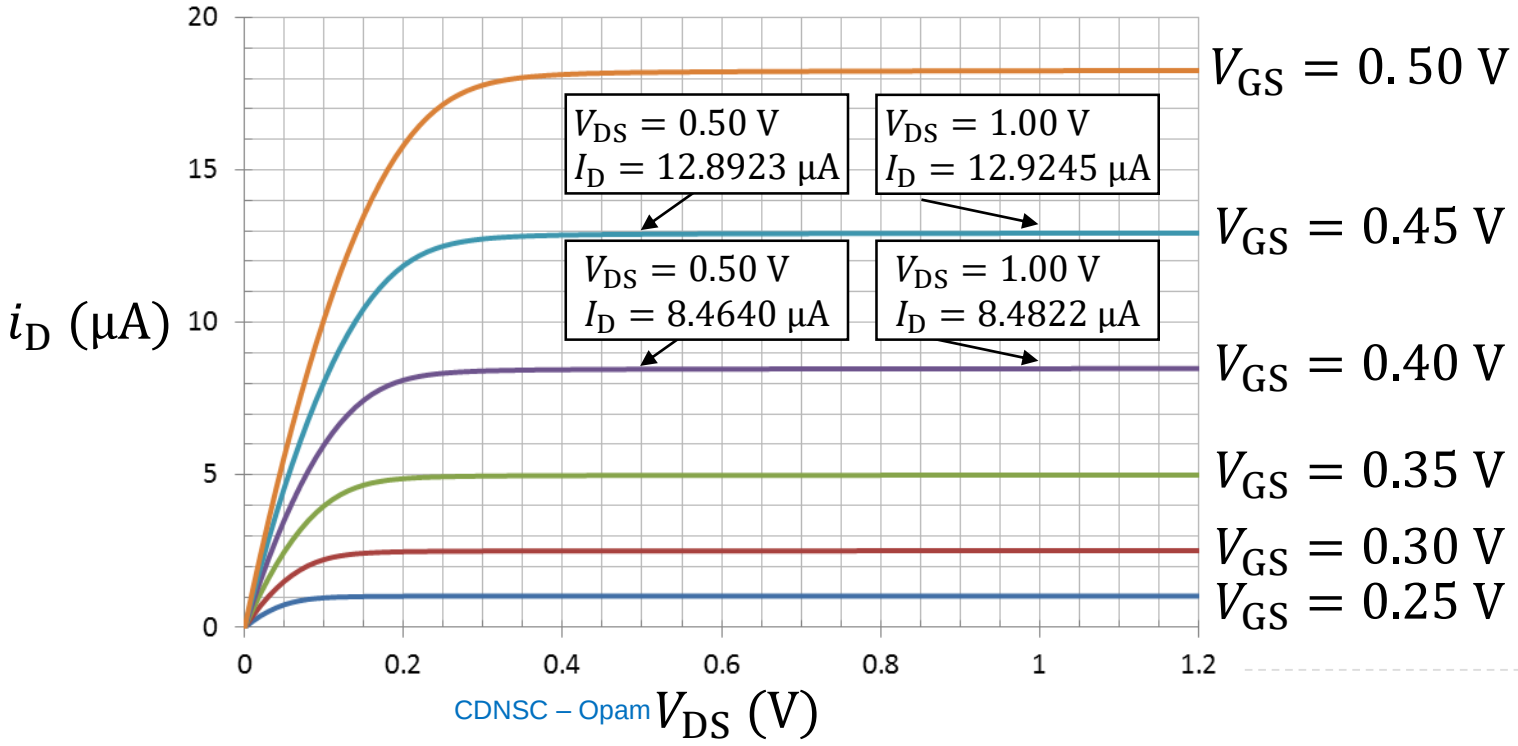
Figure 3: Measurement setup



Exercise 1-2 – Task 4.a

4. In **Figure 4**, a simulated set of transfer characteristics of an NMOS transistor is shown ($W/L = 100\ \mu m/100\ \mu m$, $\beta_0 = 360\ \mu A/V^2$)

   a. Mark and label the different operating regions in Figure 4. Describe what happens at the transition from the linear to the saturation region?

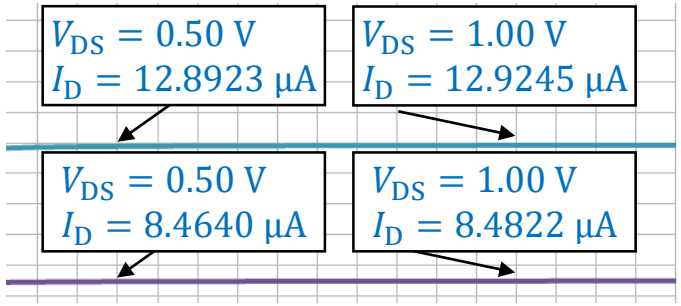


Exercise 1-2 – Task 4.b (1/3)

b. Extract the channel-length modulation factor λ using two of the four points annotated in Figure 4! Based on your findings for λ, determine the threshold voltage V_{th}!

Determination of λ

$$i_{\mathrm{D}} = \frac{\beta_0}{2}\frac{W}{L}(v_{\mathrm{GS}} - V_{\mathrm{th}})^2(1 + \lambda v_{\mathrm{DS}})$$



For a fixed v_{GS}, it thus follows

$$\frac{I_{\mathrm{D1}}}{I_{\mathrm{D2}}} = \frac{1 + \lambda V_{\mathrm{DS1}}}{1 + \lambda V_{\mathrm{DS2}}}$$

Solving for λ yields

$$\lambda = \frac{I_{\mathrm{D1}} - I_{\mathrm{D2}}}{I_{\mathrm{D2}} V_{\mathrm{DS2}} - I_{\mathrm{D1}} V_{\mathrm{DS1}}}$$

For $V_{\mathrm{GS}} = 0.40\ \mathrm{V}$

$$\lambda = \frac{8.4640\ \mu\mathrm{A} - 8.4822\ \mu\mathrm{A}}{8.4822\ \mu\mathrm{A} \cdot 0.5\ \mathrm{V} - 8.4640\ \mu\mathrm{A} \cdot 1\ \mathrm{V}} = 0.00431\ \mathrm{V}^{-1}$$

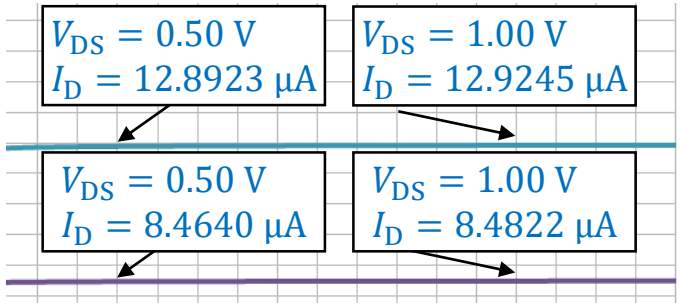
For $V_{\mathrm{GS}} = 0.45\ \mathrm{V}$

$$\lambda = 0.00501\ \mathrm{V}^{-1}$$

Exercise 1-2 – Task 4.b (2/3)

b. Extract the channel-length modulation factor λ using two of the four points annotated in Figure 4! Based on your findings for λ, determine the threshold voltage V_{th}!

Determination of V_{th}:



Neglecting channel-length modulation, V_{th} results in

$$i_{\mathrm{D}} = \frac{\beta_0}{2}\frac{W}{L}(v_{\mathrm{GS}} - V_{\mathrm{th}})^2$$

$$\rightarrow V_{\mathrm{th}} = v_{\mathrm{GS}} - \sqrt{\frac{2Li_{\mathrm{D}}}{\beta_0 W}} = 0.4\ \mathrm{V} - \sqrt{\frac{2 \cdot 100\ \mu\mathrm{m} \cdot 8.4640\ \mu\mathrm{A}}{100\ \mu\mathrm{m} \cdot 360\ \frac{\mu\mathrm{A}}{\mathrm{V}^2}}} = 183\ \mathrm{mV}$$

$V_{\mathrm{GS}} = 0.40\ \mathrm{V}$, $V_{\mathrm{DS}} = 0.5\ \mathrm{V}$, $I_{\mathrm{D}} = 8.4640\ \mu\mathrm{A}$ $\rightarrow V_{\mathrm{th}} \approx 183\ \mathrm{mV}$

$V_{\mathrm{GS}} = 0.40\ \mathrm{V}$, $V_{\mathrm{DS}} = 1.0\ \mathrm{V}$, $I_{\mathrm{D}} = 8.4822\ \mu\mathrm{A}$ $\rightarrow V_{\mathrm{th}} \approx 183\ \mathrm{mV}$

$V_{\mathrm{GS}} = 0.45\ \mathrm{V}$, $V_{\mathrm{DS}} = 0.5\ \mathrm{V}$, $I_{\mathrm{D}} = 12.8923\ \mu\mathrm{A}$ $\rightarrow V_{\mathrm{th}} \approx 182\ \mathrm{mV}$

$V_{\mathrm{GS}} = 0.45\ \mathrm{V}$, $V_{\mathrm{DS}} = 1.0\ \mathrm{V}$, $I_{\mathrm{D}} = 12.9245\ \mu\mathrm{A}$ $\rightarrow V_{\mathrm{th}} \approx 182\ \mathrm{mV}$

Exercise 1-2 – Task 4.b (3/3)

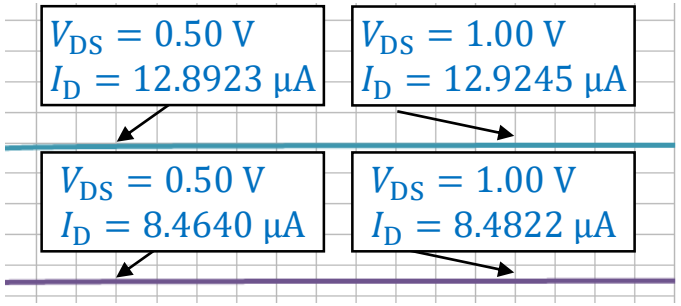
- b-Extract the channel-length modulation factor λ using two of the four points annotated in Figure 4! Based on your findings for λ, determine the threshold voltage V_{th}!

Determination of V_{th}

With channel-length modulation, V_{th} results in

$$i_{\mathrm{D}} = \frac{\beta_0}{2}\frac{W}{L}(v_{\mathrm{GS}} - V_{\mathrm{th}})^2(1 + \lambda v_{\mathrm{DS}})$$

$$\rightarrow V_{\mathrm{th}} = v_{\mathrm{GS}} - \sqrt{\frac{2Li_{\mathrm{D}}}{\beta_0 W(1 + \lambda v_{\mathrm{DS}})}}$$



$V_{\mathrm{GS}} = 0.40$ V, $V_{\mathrm{DS}} = 0.5$ V, $I_{\mathrm{D}} = 8.4640$ µA, $\lambda = 0.00431\ \mathrm{V}^{-1}$ $\rightarrow V_{\mathrm{th}} \approx 183$ mV

$V_{\mathrm{GS}} = 0.40$ V, $V_{\mathrm{DS}} = 1.0$ V, $I_{\mathrm{D}} = 8.4822$ µA, $\lambda = 0.00431\ \mathrm{V}^{-1}$ $\rightarrow V_{\mathrm{th}} \approx 183$ mV

$V_{\mathrm{GS}} = 0.45$ V, $V_{\mathrm{DS}} = 0.5$ V, $I_{\mathrm{D}} = 12.8923$ µA, $\lambda = 0.00501\ \mathrm{V}^{-1}$ $\rightarrow V_{\mathrm{th}} \approx 182$ mV

$V_{\mathrm{GS}} = 0.45$ V, $V_{\mathrm{DS}} = 1.0$ V, $I_{\mathrm{D}} = 12.9245$ µA, $\lambda = 0.00501\ \mathrm{V}^{-1}$ $\rightarrow V_{\mathrm{th}} \approx 182$ mV

Exercise 1-2 – Task 4.c

c. The transistor is to be biased as a current source that delivers a current of $3\ \mu\text{A}$. Based on your findings for V_{th} in task 4.b, <u>determine</u> the gate-source-voltage V_{GS} for

- $\lambda = 0$

$$I_{\text{D}} = \frac{\beta_0}{2}\frac{W}{L}(V_{\text{GS}} - V_{\text{th}})^2$$

$$\rightarrow V_{\text{GS}} = V_{\text{th}} + \sqrt{\frac{2LI_{\text{D}}}{\beta_0 W}} = 183\ \text{mV} + \sqrt{\frac{2 \cdot 3\ \mu\text{A} \cdot 100\ \mu\text{m}}{360\frac{\mu\text{A}}{\text{V}^2} \cdot 100\ \mu\text{m}}} \approx 312\ \text{mV}$$

- λ = “your result of task 4.b” and $V_{\text{DS}} = 1\ \text{V}$.

$$I_{\text{D}} = \frac{\beta_0}{2}\frac{W}{L}(V_{\text{GS}} - V_{\text{th}})^2\ (1 + \lambda V_{\text{DS}})$$

$$\rightarrow V_{\text{GS}} = V_{\text{th}} + \sqrt{\frac{2LI_{\text{D}}}{\beta_0 W(1 + \lambda V_{\text{DS}})}} = 183\ \text{mV} + \sqrt{\frac{2 \cdot 3\ \mu\text{A} \cdot 100\ \mu\text{m}}{360\frac{\mu\text{A}}{\text{V}^2} \cdot 100\ \mu\text{m} \cdot (1 + 0.0044\ \text{V}^{-1} \cdot 1\ \text{V})}} \approx 312\ \text{mV}$$